

YA HAKIM

Nine patients. Each hiding one thing.
The AI playing them has never been told what is wrong.

5 minute demo + 3 minute Q&A

Yasser drives

Ali narrates

Before you walk up

- Open **one** browser tab, full screen, on the **LAN address** — never localhost.
- Play one throwaway round backstage. The first AI reply of a session is slow; do not let that happen on stage.
- Patient for the demo: **Georges, 62**. Sound on. Volume up.
- Close every other tab. Notifications off.

THE FIVE MINUTES

0:00 – 0:35 THE HOOK

YASSER “We are computer science students. Three days ago neither of us could diagnose anything.”

YASSER “This is Georges. He is sixty-two, he has had headaches for six weeks, and he is hiding something. You have four minutes before he dies.”

YASSER “And the AI playing him has never been told what is wrong with him. Not told to keep a secret. **Never told.**”

► Stay on the patient screen. Do not explain the architecture yet — show the game first.

0:35 – 2:35 PLAY HIM

YASSER types, one at a time. **ALI** reads the room and narrates.

1. “is anyone else at home feeling unwell” **ALI**: “Watch the dots. That is a thread of his history.”

2. “do you feel better when you are away from the house” **ALI**: “He does not want to be here. He has an answer ready for everything.”

3. “what heats your house” **ALI**: “He was a boiler engineer for forty years.”

4. **Examine** → **Cognition** **ALI**: “His oxygen reads ninety-nine percent. **That number is lying to you**, and that is not our bug — a pulse oximeter physically cannot see this.”

► **Diagnose** → type carbon monoxide → call it. Do not wait for the clock.

ALI (as the reveal lands) “His wife and the dog are still in that house.”

► **Pause here for two seconds.** Let that sit. Do not talk over it.

2:35 – 3:50 THE PROOF — this is what wins

ALI “Every demo today will tell you their AI does something clever. We are going to show you ours cannot cheat, and you can check it yourselves.”

► Click **Show them it couldn’t cheat.**

ALI “This is the complete text we send the AI. Every word of it.”

ALI “There is a search box. Somebody give me a word.”

► Type monoxide. Then gas. Then poisoning. **Zero matches** each time.

ALI *"It is not hidden. It was never put in."*

ALI *"And above it: one hundred scripted jailbreak attacks, ten categories, including Arabic and French. Zero got through. That number is measured when you open this panel — it is not typed into the page."*

3:50 – 4:30 IT TEACHES YOU

► Go back. Click **What you should take away**.

YASSER *"After every case it tells you what mattered, and what would have **harm**ed him — here, giving him oxygen guided by that lying monitor."*

YASSER *"Every case cites the real published guidelines it came from. We are not doctors, so we did not invent the medicine."*

► Click **Next patient** — let them see all nine on the picker.

YASSER *"Nine patients, three difficulty tiers. Everything in that 3D room — the bed, the body, the monitor — is generated in code. Nothing downloaded."*

4:30 – 5:00 CLOSE

YASSER *"Ya Hakim is a game where you learn to read a patient instead of guessing."*

YASSER *"And it is built so the answer is not merely hidden from the AI — it was never given to it. That is a design decision, and we can prove it in front of you in ten seconds."*

YASSER *"One link. It is running right now. Thank you."*

THE THREE MINUTES OF QUESTIONS

Who answers what

Ali takes anything technical — architecture, the AI, the proof, the code.

Yasser takes the game, the cases, the design, and anything about what is next.

If you do not know: **say you do not know**. It beats bluffing, every time.

Q. How do you know the AI isn't just working out the diagnosis by itself?

ALI: It might. A capable model sees yellow eyes and a swollen belly and can infer liver failure — we do not claim it is stupid. We claim two things we can prove: we never gave it the answer, and if it ever says the answer, the player never sees it. Every reply is scanned before it reaches the screen, and a reply that contains the answer is thrown away and asked again — not censored, because a blanked-out word tells you the word was there.

Q. Why not just tell the model to keep it secret?

ALI: Because an instruction is negotiable. Everything in a prompt can be argued with, role-played around, or extracted. Absence cannot. There is no prompt you can write that makes a model reveal something it does not have.

Q. So how is that actually built?

ALI: An allowlist. Twelve named fields get copied **into** the prompt — name, age, personality, symptoms, the lie. Five fields are the answer and are never on that list. The obvious way round is to take the whole file and strip the secrets out, but then a field somebody adds next year leaks by default. Copying in means a new field is invisible unless you deliberately add it. It is safe when you forget.

Q. How is this different from just role-playing with ChatGPT?

ALI: The model only speaks. Everything that makes it a game — the clock, the vitals, when he cracks and tells the truth, the scoring — is our engine, and it is deterministic and tested. The model never decides anything.

Q. Is the medicine accurate?

YASSER: We are CS students, so we did not write it from memory. Every case cites published guidelines — NICE, the Resuscitation Council, the European Society of Cardiology — and they are listed at the end of each case. The vitals thresholds are NEWS2, the score the NHS actually uses at the bedside.

Q. What happens if the wifi dies?

YASSER: It keeps playing. There is a full offline patient with scripted replies, and from the back of a room you cannot tell. That was not a fallback we bolted on — it is how we built it before we had an API key.

Q. What was the hardest bug?

ALI: The guard blocks any word from the diagnosis. One diagnosis was “liver disease **from** untreated hepatitis B” — so it banned the word “from” and the patient could not speak an English sentence. We had to add a stopword list. The other one: a naive search for “liver” flags “**delivered**”, so everything matches on whole words now, and there is a test with the sentence “*my wife delivered the news*”.

Q. What would you do next?

YASSER: Three things, in order. More patients — nine is an hour of play. Then finish the examinations: about half of them return a generic normal result today. Then multiplayer, which is already built and tested but not switched on — a room of phones against one projector.

Q. Can a player cheat by looking at the network traffic?

ALI: Not during the round — the answer is in no payload the browser receives, and that is tested. One honest caveat: the proof panel itself lists the answer words, because that is how it shows you a hit count of zero. So yes, if you open developer tools mid-round and go looking for our proof endpoint, you can read them there. We would gate it if this were shipping.

Do not say these

- **Do not promise his heart spikes when he lies.** It works about half the time — it is a known bug. If it fires, great, point at it. Never set it up in advance.
- **Do not call it a medical tool.** It is a game that teaches reasoning. Saying otherwise invites a question you cannot win.
- **Do not claim the AI “cannot know”.** It was never told. That is the honest claim and it is strong enough.

If something breaks on stage

The AI is slow or silent. Keep talking, the offline patient answers instantly and nobody can tell.

The 3D room does not load. Say *“the room is optional, the game is the conversation”* and carry on. It is built to fail that way on purpose.

The page is blank. Refresh once. The round is on the server, not the browser.

You freeze. Hand to the other one. That is what two people are for.

The numbers, if you are asked

9	patients, across three difficulty tiers
12 / 5	fields the AI may see / fields that are the answer and never leave the server
100	scripted jailbreak attacks, in 10 categories, including Arabic and French
0	that got through
740	automated tests, all passing
20,375	lines of code, written in five days
0	downloaded 3D models, textures or images — the room is generated

Say less than you want to. Let the zero-matches moment breathe.